

Homework 5

Problem 1 Flapping angle $\phi(t)$ (controlled input), passive pitching angle is: $q = \psi, \dot{q} = \dot{\psi}$

Body angular velocity components $w_x = \dot{\psi}, w_y = \dot{\phi} \cos \psi, w_z = -\dot{\phi} \sin \psi \therefore w = \begin{bmatrix} \dot{\psi} \\ \dot{\phi} \cos \psi \\ -\dot{\phi} \sin \psi \end{bmatrix}$

Inertia matrix: $I = \begin{bmatrix} I_{xx} & -I_{xy} & 0 \\ -I_{xy} & I_{yy} & 0 \\ 0 & 0 & I_{xx} + I_{yy} \end{bmatrix}$, Rotational kinetic energy $T = \frac{1}{2} w^T I w$

$$I w = \begin{bmatrix} I_{xx} & -I_{xy} & 0 \\ -I_{xy} & I_{yy} & 0 \\ 0 & 0 & I_{xx} + I_{yy} \end{bmatrix} \begin{bmatrix} \dot{\psi} \\ \dot{\phi} \cos \psi \\ -\dot{\phi} \sin \psi \end{bmatrix} = \begin{bmatrix} I_{xx}(\dot{\psi}) + (-I_{xy})(\dot{\phi} \cos \psi) + 0 \\ (-I_{xy})(\dot{\psi}) + I_{yy}(\dot{\phi} \cos \psi) + 0 \\ 0 + 0 + (I_{xx} + I_{yy})(-\dot{\phi} \sin \psi) \end{bmatrix} = \begin{bmatrix} I_{xx}\dot{\psi} - I_{xy}\dot{\phi} \cos \psi \\ -I_{xy}\dot{\psi} + I_{yy}\dot{\phi} \cos \psi \\ -(I_{xx} + I_{yy})\dot{\phi} \sin \psi \end{bmatrix}$$

$$\begin{aligned} w^T I w &= \begin{bmatrix} \dot{\psi} & \dot{\phi} \cos \psi & -\dot{\phi} \sin \psi \end{bmatrix} \begin{bmatrix} I_{xx}\dot{\psi} - I_{xy}\dot{\phi} \cos \psi \\ -I_{xy}\dot{\psi} + I_{yy}\dot{\phi} \cos \psi \\ -(I_{xx} + I_{yy})\dot{\phi} \sin \psi \end{bmatrix} \\ &= I_{xx}\dot{\psi}^2 - \underline{I_{xy}\dot{\psi}\dot{\phi} \cos \psi} - \underline{I_{xy}\dot{\phi}\dot{\psi} \cos \psi} + I_{yy}\dot{\phi}^2 \cos^2 \psi + (I_{xx} + I_{yy})\dot{\phi}^2 \sin^2 \psi \\ &= I_{xx}\dot{\psi}^2 - 2I_{xy}\dot{\psi}\dot{\phi} \cos \psi + \underline{I_{yy}\dot{\phi}^2 \cos^2 \psi} + I_{xx}\dot{\phi}^2 \sin^2 \psi + \underline{I_{yy}\dot{\phi}^2 \sin^2 \psi} \quad \cos^2 \psi + \sin^2 \psi = 1 \\ &= I_{xx}\dot{\psi}^2 - 2I_{xy}\dot{\psi}\dot{\phi} \cos \psi + I_{yy}\dot{\phi}^2 + I_{xx}\dot{\phi}^2 \sin^2 \psi \end{aligned}$$

$$T = \frac{1}{2} w^T I w = \frac{1}{2} I_{xx}\dot{\psi}^2 - I_{xy}\dot{\psi}\dot{\phi} \cos \psi + \frac{1}{2} I_{yy}\dot{\phi}^2 + \frac{1}{2} I_{xx}\dot{\phi}^2 \sin^2 \psi \quad \text{Euler-Lagrange equation } \frac{d}{dt} \left(\frac{\partial T}{\partial \dot{\psi}} \right) - \frac{\partial T}{\partial \psi} = M_x$$

$$\frac{\partial T}{\partial \dot{\psi}} = I_{xx}\dot{\psi} - I_{xy}\dot{\phi} \cos \psi \quad \frac{d}{dt} \left(\frac{\partial T}{\partial \dot{\psi}} \right) = \frac{d}{dt} (I_{xx}\dot{\psi} - I_{xy}\dot{\phi} \cos \psi) = I_{xx}\ddot{\psi} + I_{xy}\dot{\phi} \sin \psi \dot{\psi} - I_{xy}\ddot{\phi} \cos \psi$$

$$\frac{\partial T}{\partial \psi} = 0 + I_{xy}\dot{\psi}\dot{\phi} \sin \psi + 0 + I_{xx}\dot{\phi}^2 \sin \psi \cos \psi = I_{xy}\dot{\psi}\dot{\phi} \sin \psi + I_{xx}\dot{\phi}^2 \sin \psi \cos \psi$$

$$\begin{aligned} \Rightarrow M_x &= I_{xx}\ddot{\psi} + I_{xy}\dot{\phi} \sin \psi \dot{\psi} - I_{xy}\ddot{\phi} \cos \psi - (I_{xy}\dot{\psi}\dot{\phi} \sin \psi + I_{xx}\dot{\phi}^2 \sin \psi \cos \psi) \\ &= I_{xx}\ddot{\psi} - I_{xy}\ddot{\phi} \cos \psi - I_{xx}\dot{\phi}^2 \sin \psi \cos \psi \\ &= I_{xx}\ddot{\psi} - I_{xy}\ddot{\phi} \cos \psi - \frac{1}{2} I_{xx}\dot{\phi}^2 \sin(2\psi) \end{aligned}$$

Trig. Identity $\sin(2\psi) = 2 \sin \psi \cos \psi$
 $\frac{1}{2} \sin(2\psi) = \sin \psi \cos \psi$

$$\therefore I_{xx}\ddot{\psi} = M_x + I_{xy}\ddot{\phi} \cos \psi + \frac{1}{2} I_{xx}\dot{\phi}^2 \sin(2\psi)$$

Problem 2 1. Turn the system of equations into four first order differential equations $\frac{d\vec{x}}{dt} = g(\vec{x})$.

$$\text{Given } I\ddot{\phi} + T|\dot{\phi}| \phi + \frac{k\phi}{T^2} = \frac{F}{T}, \ddot{F} + r_3(1+k)\dot{F} + kr_3^2 F = -\mu kr_3^2 \dot{\phi}$$

$$\text{State vector } \vec{x} = \begin{bmatrix} \phi \\ \dot{\phi} \\ F \\ \dot{F} \end{bmatrix} \quad x_1 = \phi, x_2 = \dot{\phi}, x_3 = F, x_4 = \dot{F} \quad \dot{x}_1 = x_2, \dot{x}_3 = x_4$$

$$\Rightarrow I\ddot{\phi} + T|\dot{\phi}| \phi + \frac{k\phi}{T^2} = \frac{F}{T} \quad I\ddot{\phi} = \frac{F}{T} - T|\dot{\phi}| \phi - \frac{k\phi}{T^2} \Rightarrow \ddot{\phi} = \frac{1}{I} \left(\frac{F}{T} - T|\dot{\phi}| \phi - \frac{k\phi}{T^2} \right) \Rightarrow \dot{x}_2 = \frac{1}{I} \left(\frac{x_3}{T} - T|x_2|x_2 - \frac{kx_1}{T^2} \right)$$

$$\Rightarrow \ddot{F} + r_3(1+k)\dot{F} + kr_3^2 F = -\mu kr_3^2 \dot{\phi} \Rightarrow \ddot{F} = -r_3(1+k)\dot{F} - kr_3^2 F - \mu kr_3^2 \dot{\phi} \Rightarrow \dot{x}_4 = -r_3(1+k)x_4 - kr_3^2 x_3 - \mu kr_3^2 x_2$$

$$\dot{\vec{x}} = g(\vec{x}) = \begin{bmatrix} x_2 \\ \frac{1}{I} \left(\frac{x_3}{T} - T|x_2|x_2 - \frac{kx_1}{T^2} \right) \\ x_4 \\ -r_3(1+k)x_4 - kr_3^2 x_3 - \mu kr_3^2 x_2 \end{bmatrix}$$

2. Determine the fixed point(s) of this system, $g(\vec{x}^*) = 0$ $\dot{\vec{x}} = 0, \dot{x}_1 = x_2 = 0, x_2^* = 0, \dot{x}_3 = x_4 = 0, x_4^* = 0$

$$\dot{x}_2 = 0 = \frac{1}{I} \left(\frac{x_3}{T} - T|x_2|x_2 - \frac{kx_1}{T^2} \right) \Rightarrow 0 = \frac{x_3}{T} - \frac{kx_1}{T^2} \quad T x_3 = k x_1 \quad 0 = k x_1 \quad x_1^* = 0$$

$$\dot{x}_4 = 0 = -r_3(1+k)x_4 - kr_3^2 x_3 - \mu kr_3^2 x_2 = -kr_3^2 x_3 \quad \therefore x_3^* = 0 \quad \therefore \text{the fixed point is } \vec{x}^* = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

3. Linearize the system about the fixed point, $\frac{d\vec{x}}{dt} \approx \frac{d}{d\vec{x}} g(\vec{x})|_{\vec{x}=\vec{x}^*} (\vec{x} - \vec{x}^*) \rightarrow A\vec{x}$

$$A = \frac{dg}{d\vec{x}} \Big|_{\vec{x}=\vec{x}^*}$$

$$A = \begin{bmatrix} \frac{\partial g_1}{\partial x_1} & \frac{\partial g_1}{\partial x_2} & \frac{\partial g_1}{\partial x_3} & \frac{\partial g_1}{\partial x_4} \\ \frac{\partial g_2}{\partial x_1} & \frac{\partial g_2}{\partial x_2} & \frac{\partial g_2}{\partial x_3} & \frac{\partial g_2}{\partial x_4} \\ \frac{\partial g_3}{\partial x_1} & \frac{\partial g_3}{\partial x_2} & \frac{\partial g_3}{\partial x_3} & \frac{\partial g_3}{\partial x_4} \\ \frac{\partial g_4}{\partial x_1} & \frac{\partial g_4}{\partial x_2} & \frac{\partial g_4}{\partial x_3} & \frac{\partial g_4}{\partial x_4} \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ \frac{-k}{IT^2} & 0 & \frac{1}{IT} & 0 \\ 0 & 0 & 0 & 1 \\ 0 & -\mu k r_3^2 & -k r_3^2 & -r_3(1+k) \end{bmatrix}$$

$\otimes \vec{x} \cdot \vec{x}^*$

4. Construct the characteristic equation for the matrix A . The characteristic equation results from expanding the equation, $\det(A - \lambda I) = 0$, into a fourth order polynomial in λ with four possible eigenvalue solutions $\lambda_{i=1, \dots, 4}$. (Note I is the identity matrix).

$\det(A - \lambda I) = 0$ Look a python code

```
In [1]: import sympy as sp
import numpy as np
import matplotlib.pyplot as plt
```

Part 4

```
In [8]: # Symbols
lam = sp.symbols('lambda')
I, Gamma, k, T = sp.symbols('I Gamma k T', positive=True)
mu, kappa, r3 = sp.symbols('mu kappa r3', positive=True)

# Linearized A matrix
A = sp.Matrix([
    [0, 1, 0, 0],
    [-k/(I*T**2), 0, 1/(I*T), 0],
    [0, 0, 0, 1],
    [0, -mu*kappa*r3**2, -kappa*r3**2, -r3*(1+kappa)]
])

# Characteristic equation det(A - Lambda*I) = 0
char_poly = sp.factor((A - lam*sp.eye(4)).det())
sp.simplify(char_poly)
char_poly
```

```
Out[8]: 
$$\frac{IT^2\kappa\lambda^3r_3 + IT^2\kappa\lambda^2r_3^2 + IT^2\lambda^4 + IT^2\lambda^3r_3 + T\kappa\lambda\mu r_3^2 + k\kappa\lambda r_3 + k\kappa r_3^2 + k\lambda^2 + k\lambda r_3}{IT^2}$$

```

Part 5

```
In [10]: omega = sp.symbols('omega', real=True)

# Use positive-leading characteristic equation
poly = (
    I*T**2*lam**4
    + I*T**2*r3*(1+kappa)*lam**3
    + (I*T**2*kappa*r3**2 + k)*lam**2
    + (k*r3*(1+kappa) + T*mu*kappa*r3**2)*lam
    + k*kappa*r3**2
)

# Substitute Lambda = i omega
poly_iw = sp.expand(poly.subs(lam, sp.I*omega))

real_eq = sp.simplify(sp.re(poly_iw))
imag_eq = sp.simplify(sp.im(poly_iw))

real_eq
```

```
Out[10]: 
$$-IT^2\kappa\omega^2r_3^2 + IT^2\omega^4 + k\kappa r_3^2 - k\omega^2$$

```

In [11]: `imag_eq`

Out[11]: $\omega r_3 (-IT^2\kappa\omega^2 - IT^2\omega^2 + T\kappa\mu r_3 + k\kappa + k)$

Part 6

```
In [18]: import numpy as np
import matplotlib.pyplot as plt

# Fixed parameters
I_val = 1.0
T_val = 1.0
k_val = 1.0
kappa_val = 0.5

r3_vals = np.linspace(0.01, 5.0, 1000)

# Actual nontrivial boundary:
# mu = (1+kappa)(I*T^2*kappa*r3^2 - k)/(T*kappa*r3)

mu_boundary = (
    (1 + kappa_val)
    * (I_val*T_val**2*kappa_val*r3_vals**2 - k_val)
    / (T_val*kappa_val*r3_vals)
)

# Keep only mu > 0
mask = mu_boundary > 0

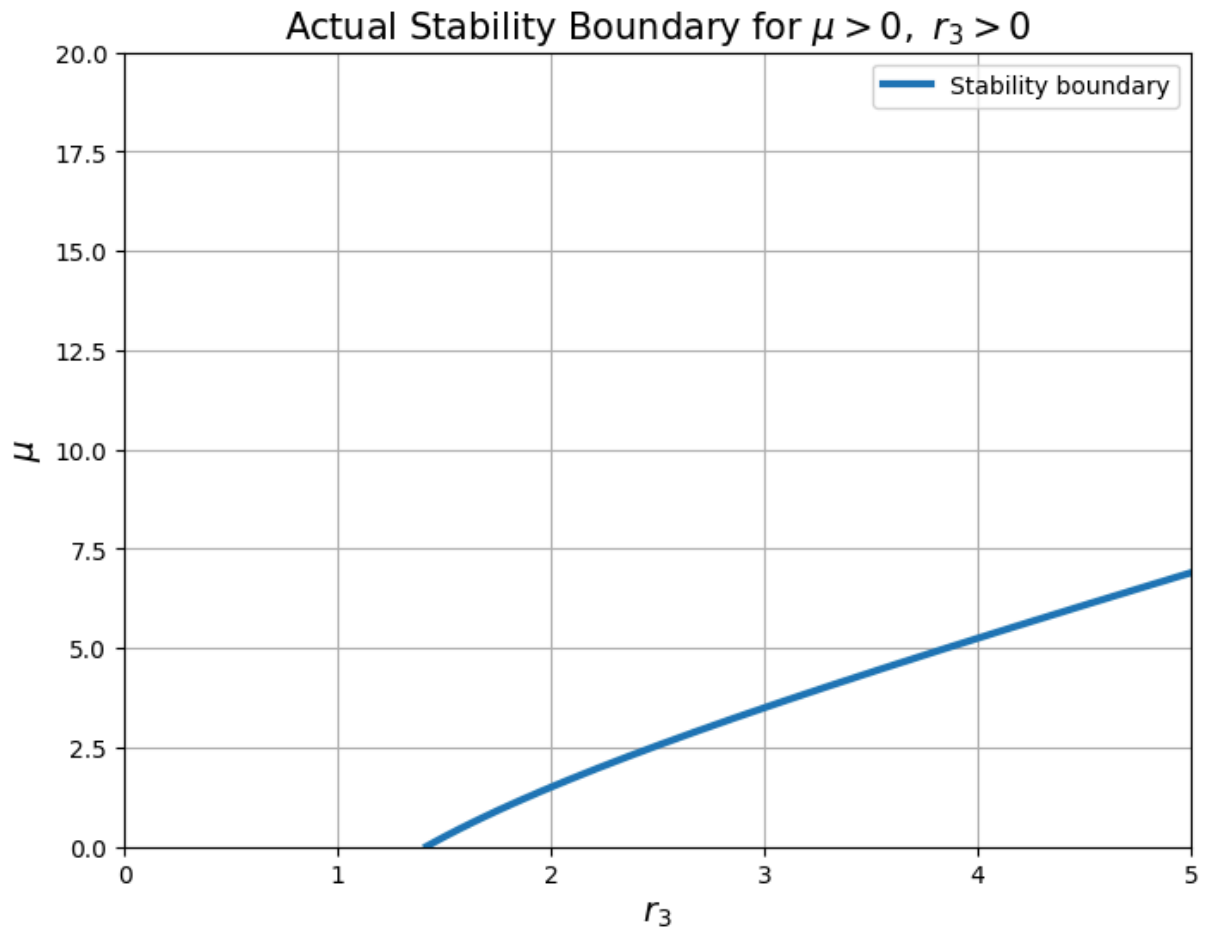
plt.figure(figsize=(8,6))

plt.plot(
    r3_vals[mask],
    mu_boundary[mask],
    linewidth=3,
    label="Stability boundary"
)

plt.xlabel(r"$r_3$", fontsize=14)
plt.ylabel(r"$\mu$", fontsize=14)
plt.title(r"Actual Stability Boundary for $\mu>0, \ r_3>0$", fontsize=15)

plt.xlim(0, 5)
plt.ylim(0, 20)

plt.grid(True)
plt.legend()
plt.show()
```



Part 7

```
In [14]: # Parameter grid
r3_vals = np.linspace(0.01, 5.0, 300)
mu_vals = np.linspace(0.01, 20.0, 300)

stability = np.zeros((len(mu_vals), len(r3_vals)))

for i, mu_val in enumerate(mu_vals):
    for j, r3_val in enumerate(r3_vals):

        A_num = np.array([
            [0, 1, 0, 0],
            [-k_val/(I_val*T_val**2), 0, 1/(I_val*T_val), 0],
            [0, 0, 0, 1],
            [0, -mu_val*kappa_val*r3_val**2,
             -kappa_val*r3_val**2,
             -r3_val*(1+kappa_val)]
        ], dtype=float)

        eigvals = np.linalg.eigvals(A_num)

        # Red if at least one eigenvalue has positive real part
        if np.any(np.real(eigvals) > 0):
            stability[i, j] = 1
```

```

        else:
            stability[i, j] = 0

plt.figure(figsize=(8, 6))

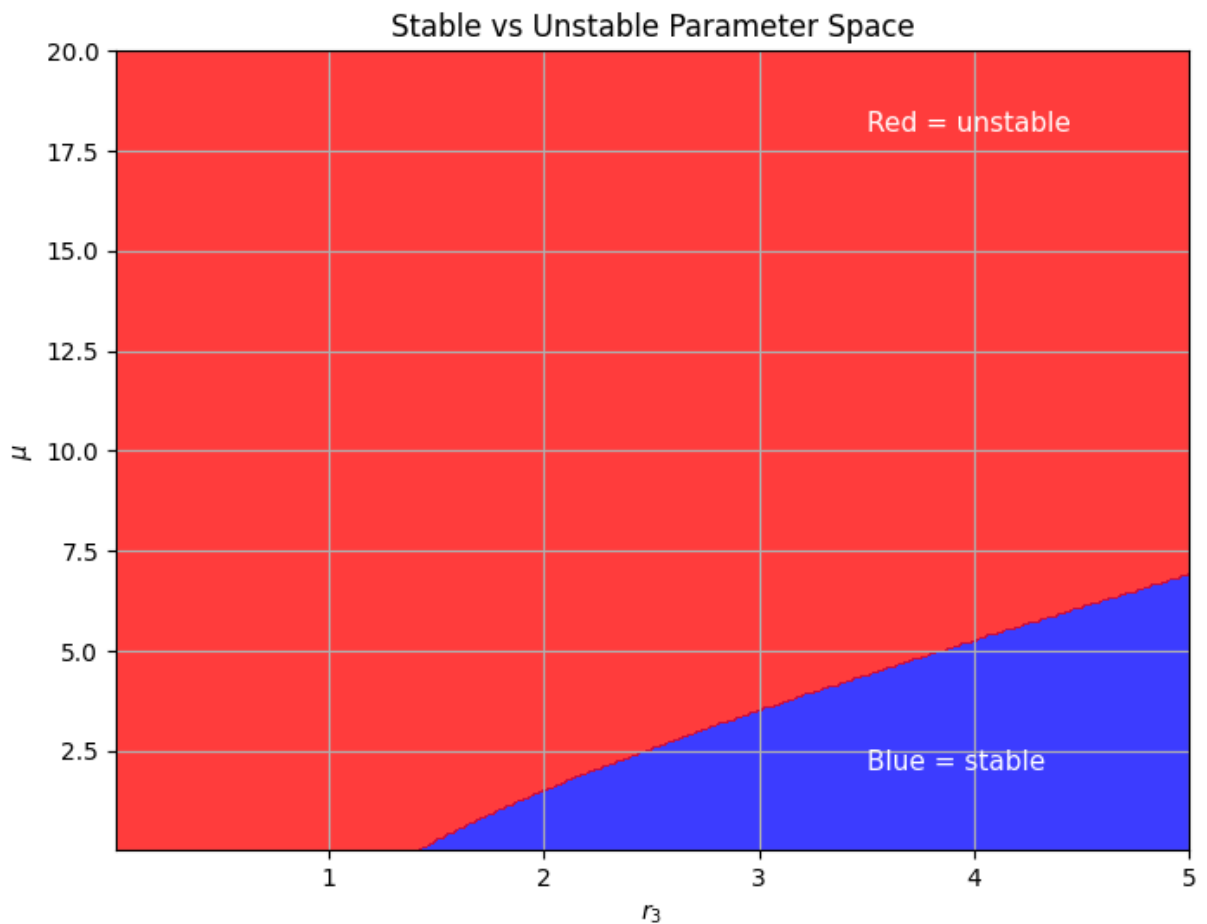
plt.contourf(
    r3_vals,
    mu_vals,
    stability,
    levels=[-0.5, 0.5, 1.5],
    colors=["blue", "red"],
    alpha=0.75
)

plt.xlabel(r"$r_3$")
plt.ylabel(r"$\mu$")
plt.title("Stable vs Unstable Parameter Space")
plt.grid(True)

plt.text(3.5, 2, "Blue = stable", color="white", fontsize=11)
plt.text(3.5, 18, "Red = unstable", color="white", fontsize=11)

plt.show()

```



In []: